

Part 1 :- Feature Selection [Filter Based Technique]

Feature Selection is the process of choosing the most relevant input features for a machine learning and removing irrelevant or redundant ones.

Why Feature Selection is important :-

i) Improved Model Accuracy :-

- Not all features help to predict the target
- Some features add noise (random info)
- Noise confuses the model.

Without feature selection

- model learns patterns from irrelevant features.
- Predictions become less accurate.

With feature selection

- model focuses only on important signals
- improves accuracy, learns cleaner relationships.

ii) Reduces Overfitting :-

- Overfitting happens when a model learns training data too well, including noise.
- More features $\Rightarrow$ more chance to fit noise.

without

- model memorizes irrelevant patterns
- Performs well on training data and performs poorly on new data.

with

- Fewer meaning features
- model learns general pattern
- more features $\neq$ better model.

iii) Reduces Multicollinearity :-

- when two or more features are highly correlated

The Problem :-

- Co-efficients become unstable
- hard to interpret feature importance
- more predictions become unreliable

Feature Selection helps

- removing one of the correlated feature
- making coefficient stable.
- improves model reliability

iv) Faster Training Time:-

- more features = more calculations
- High dimensional data slows training

Ex:- 1000 features $\rightarrow$ slow
50 important features $\rightarrow$ fast

v) Improves model Interpretability:-

- models with fewer features are easier to understand
- Especially important in : finance, healthcare, Business decisions

* vi) Reduces Curse of Dimensionality:-

$\rightarrow$ When a number of features increases, the data becomes sparse and distance-based models perform poorly

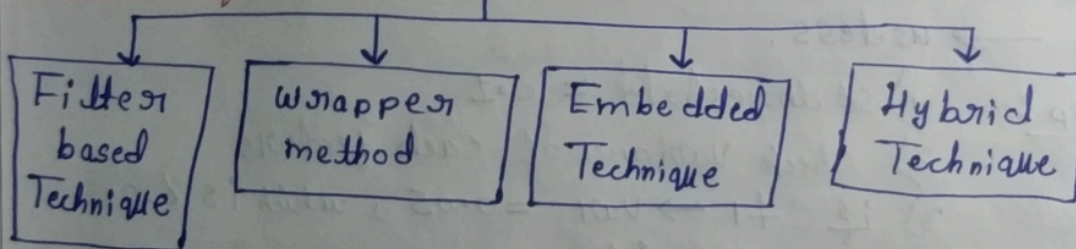
- affected algorithms:- KNN, k-means, SVM

$\rightarrow$ Feature Selection helps by

- Reducing dimensions
- making distance calculation meaningful
- improving model performance

vii) Saves cost

Types of Feature Selection:-



i) Filter methods in Feature Selection:-

Filter methods select features before training the machine learning model.

- They use statistical techniques to measure the relationship b/w input features and the target variable.
- They filter out irrelevant features.
- Done as a Preprocessing step
- works the same for any ML model.

Common Filter Techniques :-

I) Duplicate Features :- Removing duplicate Features

f ₁	f ₂	f ₃	f ₄	f ₅	output
1	2	1	2	3	Y
2	1	2	2	3	N
3	3	3	2	3	N

Key → original cols

Values → duplicate col

{ dict → keep keys, delete values }

Same

II) Variance Threshold :-

Constant

A	B	C
1	1	Y
2	1	N
3	1	Y

{ here B is not
Explaining the
variation in C, because
it is constant
(drop)

$$\text{So } \text{var}(B) = 0$$

(no variation)

Quasi Constant

1000 → rows

↓
995 rows → 1

5 rows → 0

here $\text{Var} \approx 0$ is called
quasi Constant

1 2 3
A (variation)

• Features with very low variance carry little information

• If a feature has almost the same value for all data points → useless.

how to drop :- 1) set threshold = 0.1

2) check variance of each feature

3) if $f_1 \rightarrow \text{var} = 0.05$, which is < 0.1
then drop it

Points to Remember Before Dropping Columns :-

Variance ⇒ how much a feature changes

→ If feature is constant means, it won't affect the result
and if $\text{variance} = 0 \rightarrow \text{always drop}$

i) Choose Threshold Carefully :-

• Threshold decides how much variance is

→ Small value like $[0.01] \Rightarrow \text{removes near-constant feature}$

Keep threshold values b/w 0.1 to 0.01

Scale matters

→ Variance depends on Feature scale (0...∞)

→ Features with small numeric ranges may look unimportant.

Ex:- Feature A : range 0-1 → low variance (Exp)

Feature B : range 0-1000 → high variance (Salary)

→ Variance comparison becomes misleading without scaling.

→ Standardize or Normalize data before applying Variance threshold.

(Encoded)
iii) Don't apply on Categorical Features Directly :-

→ One hot-encoded features often have low variance

→ But they may still be important.

Ex:- is-fraud = 1 ⇒ appears rarely → low variance

• But very important for Prediction

→ So don't ^{drop} blindly ⇒ use domain understanding

iv) Low Variance ≠ useless :- a low variance feature can still be predictive. Ex:- medical test ⇒ rare - abnormal values

v) Apply only on Features (NOT Target) :- ^{input}

• Variance threshold is used only on the features

• Never apply it to the target variable

vi) Use as a first Filtering Step :-

• Variance Threshold is a basic filter:

• It should be used before

• correlation analysis

• VIF

• model-based selection

vii) Check data type before dropping

viii) Always check impact of model performance, after dropping columns.

III Correlation :-

Correlation measures how strongly two variables are correlated.

→ Range :- -1 to $+1$

→ $+1 \Rightarrow$ strong positive relationship

→ $-1 \Rightarrow$ strong -ve relationship

→ $0 \Rightarrow$ no relationship.

Types of correlation :-

- i) Pearson ^{Commonly used} → linear relationship, numerical data
- ii) Spearman → monotonic relationship, rank-based
- iii) Kendall → small dataset, rank correlation

Two ways of Correlation used in Feature Selection :-

i) Feature $\longleftrightarrow$ Target Correlation

→ Find features strongly related to target

- if high correlation $\Rightarrow$ important feature
- if low correlation $\Rightarrow$ weak predictor

ii) Feature $\longleftrightarrow$ Feature Correlation

• Goal :- highly related features

• Highly correlated features carry same information, cause multicollinearity

Ex:- $f_1: \text{area} - \text{sqft}$ } $\text{corr}() = 0.99 \Rightarrow$ So keep only one
 $f_2: \text{area} - \text{sq-meter}$

Considerations while dropping columns :-

i) Use absolute Correlation Value :-

→ Consider $|\text{correlation}|$, strong negative correlation is also useful. Ex:- correlation = -0.85 - important feature

ii) Set a proper Threshold :-

• $|\text{corr}| > 0.8 \Rightarrow$ Very high

• $|\text{corr}| > 0.9 \Rightarrow$ Extreme

• Don't use low thresholds $\Rightarrow$ will drop too many features

iii) Never drop both correlated features

iv) Correlation $\neq$ Causation (*)

- High correlation does not mean one causes the other
- used only to reduce redundancy.

U-Shape $\Rightarrow$ Correlation

v) Works only for linear Relationships :- $\rightarrow$ If relationship exists

Correlation fails to detect non-linear relationships.

vi) Not Suitable for Categorical Variables :-

Pearson correlation is for numerical features

IV ANOVA :-

Anova (Analysis of Variance) is a filter method used to check whether a numerical feature has a significant relationship with a categorical target.

$\rightarrow$ In feature selection, we mostly use ANOVA F-test

When to use ANOVA :-

$\rightarrow$ Feature $\Rightarrow$ Numerical (continuous)

$\rightarrow$ Target $\Rightarrow$ Categorical (classification problem)

ANOVA Checks : Do the mean values of a feature differ significantly across different target classes?

- if class means are very different $\Rightarrow$ feature is important
- if class means are similar $\Rightarrow$ feature is not useful

ANOVA Compares two types of Variance :-

i) Between-Class :- How much the class means differ from each other

ii) Within-Class :- How much data varies inside the class

$$\text{ANOVA F-Score} = \frac{\text{Between-class Variance}}{\text{Within-class Variance}}$$

Key Considerations while applying ANOVA:-

1) Assumption of Normality:-

- ANOVA assumes that data in each class follows a normal distribution (bell-shaped).
 - Real-world data is often skewed
- ⇒ impact: if data is highly skewed → F-score may be unreliable
- Feature importance may be misleading.

2) Assumption of Homogeneity of Variance (Equal Variance)

- ANOVA assumes that all classes have similar variances
Ex class A spread $\approx$ class B spread

→ In reality, one class may be much more spread out
Ex:- churned vs non churned customers

→ Unequal variance ⇒ incorrect F-statistic

3) Independence of observations:-

- Each data point should be independent of others.
 - ANOVA is not suitable for time-series feature selection
- ⇒ time series data (today depends on yesterday)

4) Sensitive to outliers:-

- ANOVA relies on mean values:- One Extreme value can shift the mean & leads to bad f-score
- a useful feature may look weak
- a useless feature may look important
- ⇒ Always handle outliers before using ANOVA

5) Does Not Consider feature Interactions:-

- ANOVA Evaluates one feature at a time
- Some features are useful only when combined
- ANOVA cannot capture this

The main Limitation of ANOVA are its statistical assumptions
Sensitivity to outliers & inability to capture feature interactions.

V Chi-Square :-

The Chi-Square test is used to measure whether a categorical feature is independent or dependent on a categorical target.

→ Chi-Square Checks: Is the observed frequency different from what we would expect, if feature and target were independent?

- Big difference → strong relationship
- Small difference → weak relationship

How Chi-Square works :-

- Create Contingency table
- Calculate: $\frac{\text{observed frequency}}{\text{Expected frequency}}$

iii) Compare them

$$\chi^2 = \sum \frac{(\text{observed} - \text{Expected})^2}{\text{Expected}}$$

→ High $\chi^2 \Rightarrow$ Strong dependency → keep feature

→ Low $\chi^2 \Rightarrow$ weak dependency → drop feature

Example

Sex → Survived

	0	1
M	193	355
F	120	221

observed

	0	1
M	468	109
F	81	283

Expected

crazy

feature is important

$$\text{chi-square} \Rightarrow \frac{(468-193)^2}{193} + \frac{(109-355)^2}{355} + \frac{(81-120)^2}{120} + \frac{(283-221)^2}{221}$$

Key Considerations

i) Feature and Target must be Categorical

ii) Feature Values must be Non-Negative :-

- Chi-square cannot handle negative values std scaler
- after encoding, ensure all values ≥ 0 , Prefer Minmax Scaler, not

iii) Use P-value to Decide drop or keep

- $P \geq 0.05$ drop feature
- $P < 0.05$ keep feature

v) High $\chi^2 \neq$ Strong Prediction

• Chi-Square shows dependency, not prediction power

vi) Does not capture feature interaction.

• Tests one feature at a time

• A feature may be weak alone but strong with others.

vii) Large Dataset Bias:

• with large dataset, even weak relations look significant

VI Mutual Information :-

Mutual Information measures how much information a feature gives about the target.

→ Detects linear & non-linear relationships

• works for classification and regression

• $MI = 0 \rightarrow$ feature and target are unrelated

• Higher MI $\rightarrow$ feature is more powerful.

<u>Feature</u>	<u>Target</u>	<u>MI Function</u>
Numerical / Categorical	Categorical	Mutual-info- Classif
Numerical / Categorical	Numerical	Mutual-info- regression

Advantages of Filter Methods :-

i) Fast and Computationally Efficient :-

• No model training involved

• works well with large datasets

/ Speed, Scalability
| because of feature independence

ii) Model Independent :-

• can be used with any ML algo, No dependency on model type

iii) Reduces Overfitting :-

• Removes irrelevant and noisy features Early

iv) Simple to understand and Implement :-

• Based on statistics (Correlation, ANOVA, chi-square)

v) Good as first filtering step :- / Pre-Processing step

• Removes obvious useless features & makes further feature selection easy

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Disadvantages:-

- i) ignores feature Interaction:-
 - Evaluates one feature at a time
 - Cannot capture combined effect of features
- ii) Lower accuracy compared to wrapper / Embedded:-
 - May remove features that are useful in combination
- iii) Based on statistical Assumptions:-
 - ANOVA assumes normality & Equal Variance (homoscedasticity)
 - Chi-Square needs non-negative categorical data
- iv) Cannot Capture Non Linear Relationship:-
 - Correlation & ANOVA fail for Non-linear patterns
 - (Except Mutual Information)
- v) Threshold Selection is Manual:-
 - choosing values can be tricky
 - poor threshold $\rightarrow$ wrong feature removal.